

Histogram Processing

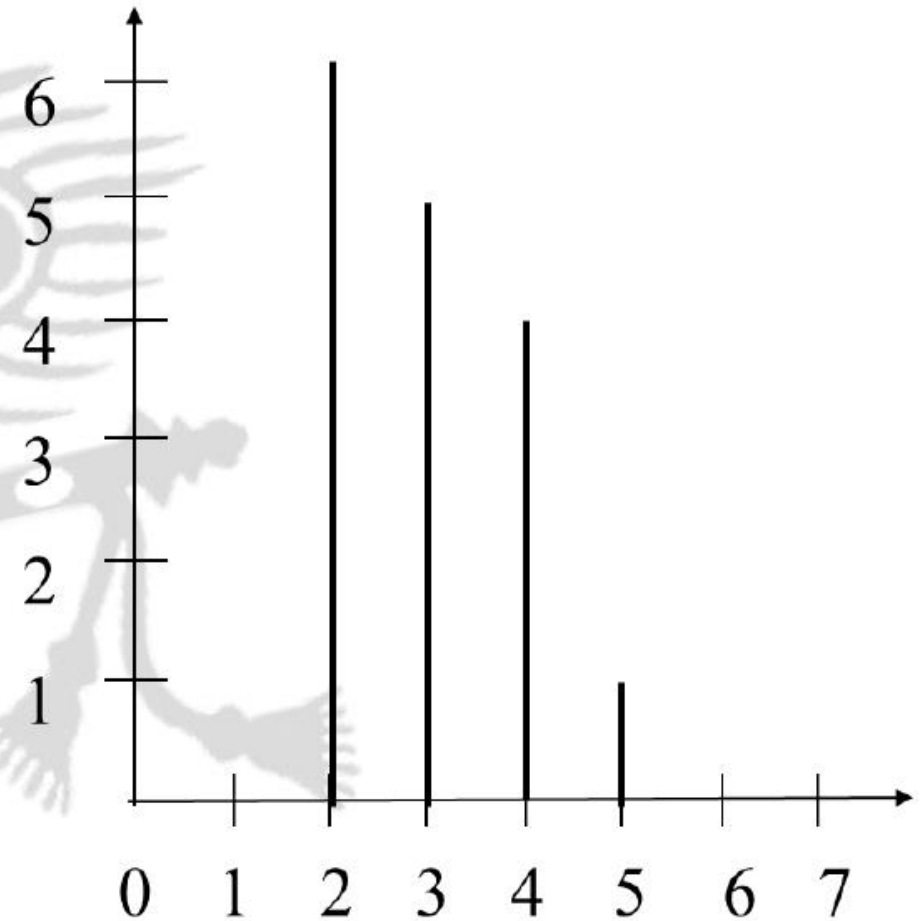
- Histogram is a discrete function formed by counting the number of pixels that have a certain gray level in the image
- Histogram of a digital image with gray levels in the range $[0, L-1]$ is a discrete function

$$h(r_k) = n_k$$

- where
- r_k : the k^{th} gray level
- n_k : the number of pixels in the image having gray level r_k
- $h(r_k)$: histogram of a digital image with gray levels r_k

Histogram Processing

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4



Normalized Histogram

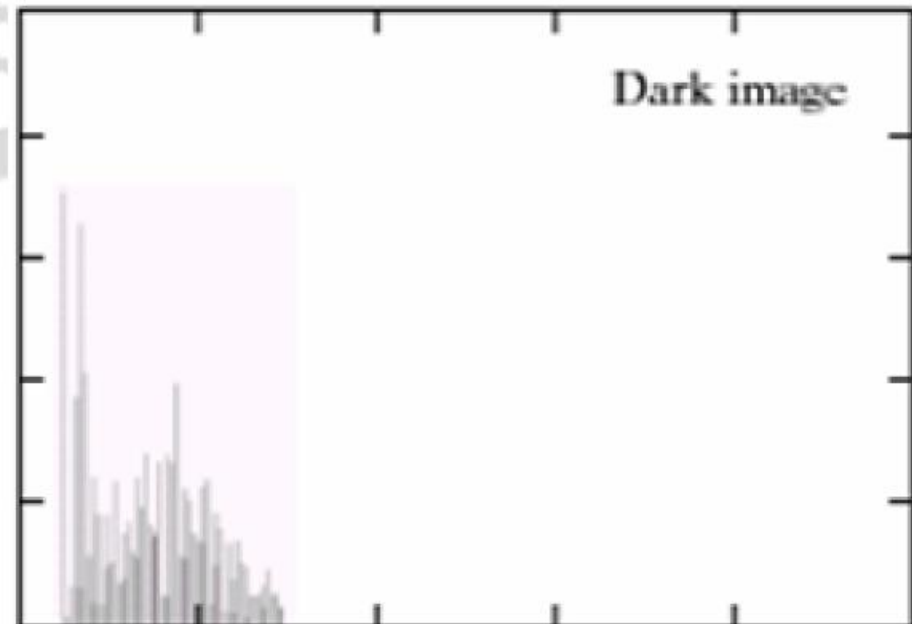
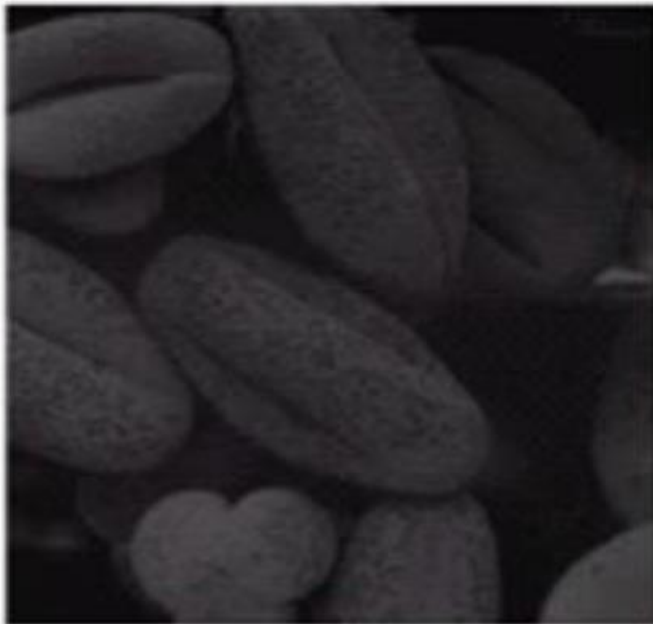
- Dividing each of histogram at gray level (r_k) by the total number of pixels in the image (n)

$$p(r_k) = n_k / n$$

- for $k = 0, 1, \dots, L-1$
- $p(r_k)$ gives an estimate of the probability of occurrence of gray level r_k
- The sum of all components of a normalized histogram is equal to 1

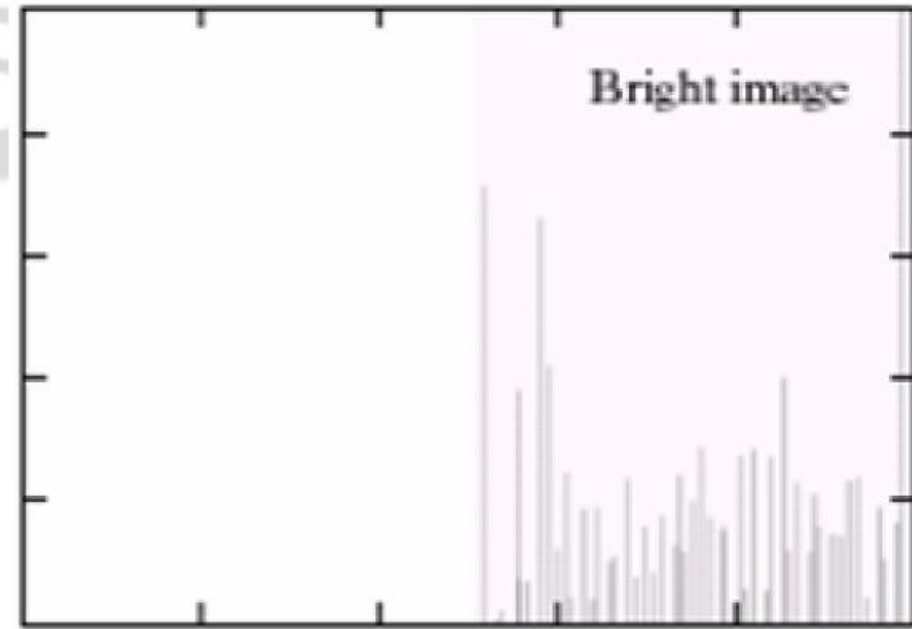
Histogram Processing

- In the dark images, components of the histogram are concentrated on the lower (dark) side of the gray scale



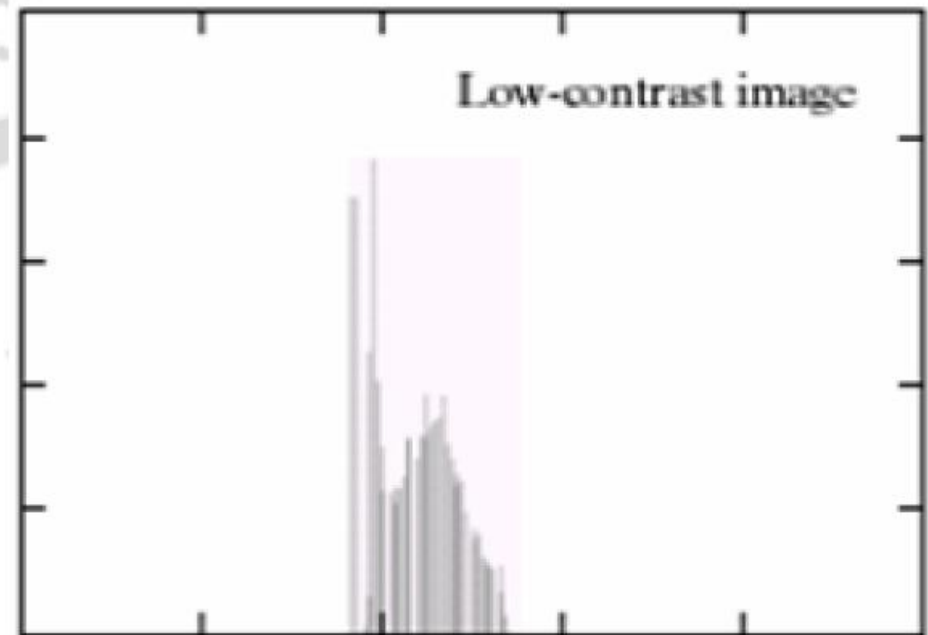
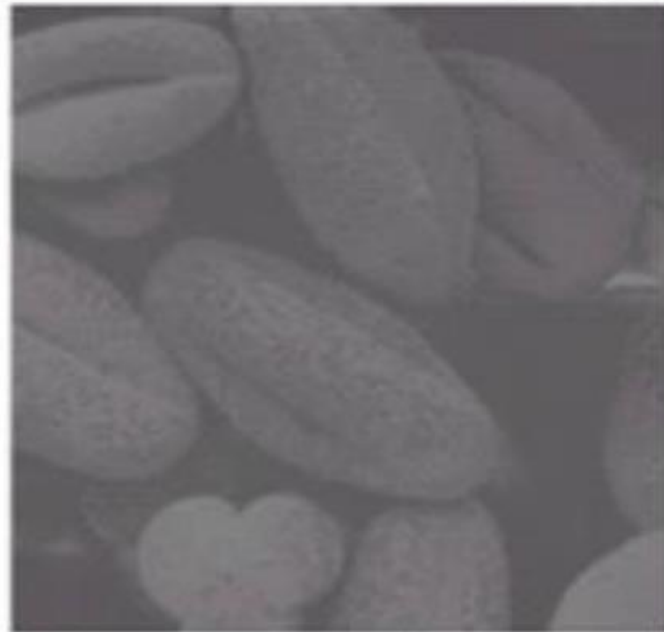
Histogram Processing

- In the dark images, components of the histogram are concentrated on the lower (dark) side of the gray scale
- In bright images, the histogram is biased towards the higher side of the gray scale



Histogram Processing

- In low contrast, the histogram will be narrow & centred towards the middle of the gray scale



Histogram Processing

- In low contrast, the histogram will be narrow & centred towards the middle of the gray scale
- In high contrast, a large variety of gray tones occupy the entire range of possible gray levels

